2021 4th International Conference on Computing and Communications Technologies (ICCCT) | 978-1-6654-1447-0/21/$31.00 ©2021 IEEE | DOI: 10.1109/ICCCT53315.2021.9711901

IoT based Smart Beekeeping Monitoring system for beekeepers in India

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***Abstract*— Beekeeping is one of the prevalent and traditional fields in agriculture, where Internet of Things (IoT)-based solutions and machine learning approaches can ease and improve beehive management significantly. A particularly important activity is bee swarming. A beehive monitoring system can be applied for digital farming to alert the user via a service about the beginning of swarming, which requires a response. An IoT-based bee activity acoustic classification system is proposed in this paper. We investigated the correlation between the temperature and humidity inside a beehive and the weight of honey produced in the hive that contains honeybees of the type Apis cerana. We measured the temperature, humidity, and weight of the beehive, using an IoT sensor. We collected data between March and April 2019 at our project site at Harali in Osmanabad, Maharashtra. We analysed the problem related to temperature and humidity to understand the effect on the bees and the overall yield of the beehive by plotting a few scatter plots. We discussed the problems which are faced by beekeepers in India and how our analysis can help them get their work done with ease.**

Keywords— Honey Bees, Beekeeping, IoT, Microcontroller, EDA, Automation.

I. Introduction

A quarter of the crops in the world, especially the orchards and floral biodiversity, requires pollination to set seeds and fruits. Bees pollinate the great majority of them. Most of the food crops need insect (mainly bee) pollinators for successful pollination in India. Oilseeds (such as Sunflower, safflower), vegetables (Cucurbitaceous Vegetable Crops, legume crops), and many fruit crops are profoundly reliant on bees.[1] In addition to their role as pollinators, these bees also produce honey and other products such as beeswax, propolis, flower pollen, bee pollen, and royal jelly which motivates farmers to collect bees. These lead to a whole new concept of farming called apiculture. Unfortunately, due to inefficient beekeeping practices, the overall yield is pretty low.

The modern ways of beekeeping are very crucial in developing nations like In-dia. There the requirement for food crops as well as bee products is very high. Most small-scale farmers are not aware of these modern practices.[2] They are still using the traditional ways of burning honeycomb to collect honey. The traditional way leads to a reduction of these pollinators in the ecosystem. Few farmers have started adopting the modern beekeeping concept by buying the beehives and taking them from one site to another according to the requirement of pollination of food crops. In recent times as the temperature is rising, the farmers have come up with local solutions to protect bees in beehives from high temperatures to help them collect more nectar. This is because the predictions made by the beekeepers tend to have human errors.

II. State of Indian Beekeeping

Based on our research, we gathered information on beekeeping practiced in various states of India. There is a huge diversity in climate, topology, and flora in India. Due to this, we found out that beekeeping culture is diversified indefinitely in various parts of India. Hence to solve this disparity and to maximize utilization of modern beekeeping practices, we have proposed a data analysis model to help beekeepers take the right decisions and increase their yield. It will motivate the beekeepers and help increase the number of beekeeping societies in various zones.

A. Beekeeping Statistics

The National Bee Board(NBB) published the current scenario of registered beekeepers till 31st December 2019. Based on the statistics, Punjab and Uttar Pradesh are leading with 271135 and 424242 bee colonies, respectively. But in large states like Maharashtra, Tamil Nadu, and Kerala the ratio of bee colonies with the geographic area is pretty less. Natural topology and flora might be favourable for practicing bee-keeping in these states. But unpredictable rainfall and other climatic conditions ren-der the beekeepers act indecisively. These lead to mismanagement of these farms. This data served as a base for us to brainstorm on various other factors restricting the growth of beekeeping in such places.[3]

B. Types of Bees in India

There are six types of bees found all over India, out of which two are local and capable of producing large amounts of honey for commercial purposes. They are known as Indian rock bees (Apis dorsata) and Apis cerana. The Indian rock bees are large(17-20mm). But they are rarely preferred for beekeeping due to their aggressive and wild nature. Moreover, rock bees prefer to live in open areas. They build honey-combs on trees. Due to this, they are difficult to rear. Hence in India, the only species that is useful in bee farming is Apis Cerana. These are smaller bees(10-11mm). They are reared by local beekeepers for commercial purposes.[4] The European bees (Apis Mellifera) imported in India, produce 75% of India's honey. When it comes to yield, they give better results than the species found locally. Both Apis Cerana and Apis Mellifera prefer to make honeycomb in dark places and small brackets. They make honeycombs in old wooden boxes. So in beehives, as we use wooden frames and as the beehives are closed, it acts as a place for these bees to make their honeycombs.

C. *Conditions for optimum Beekeeping*

The temperature of the brood nest is of extreme importance to the colony.[5] Utmost precision is necessary for maintaining it. Honey Bees maintain the brood nest temperature in the range of 32°C to 35°C so that the brood develops well. Observations suggest that Apis Cerana maintains its internal temperature between 32°C and 36°C. The average thoracic temperature of Apis Mellifera reaches about 35.7 °C in spring and 29.3 °C in summer. Both these species of bees have a similar temperature range which makes their problems somewhat the same. If the brood nest temperature crosses a threshold value, the bees start cooling the hive by fanning the hot air outside to maintain their internal conditions. Hence, providing ventilation within the nest. They also evaporate water content in honey by keeping only 20% water content inside the honey. This process helps to reduce the temperature. As a result of this, the honey becomes viscous. These evaporative cooling mechanisms protect beehives from external changes in the environment. Correspondingly, when the temperature is too low, bees generate metabolic heat by contracting and relaxing their flight muscles (having uncoupled the wings from them). The resulting vibration radiates heat in those muscles. They thermo-regulate the environment for influencing the development of brood. These processes also maintain the health of adult bees. In India, as we don't have cold weather, bees don't have to spend time generating metabolic heat. Due to the tropical climate of India, bees require ventilation. As a result of this, the yield produced reduces. This happens because bees spend more time on fanning instead of collecting nectar for honey production. Other than this, as the bee colony increases, it becomes more congested inside the hive and leads to less air circulation. This increases the humidity inside the beehive and leads to unfavorable conditions for bees to survive. The beekeeper's curiosity to check the yield of the beehive leads to disturbance in the colony. There have been cases of bee biting too. In some cases, bees also migrate away if the beekeeper checks the beehive frequently.

III. Data Analysis

A. *Data collection*

The data was collected for 2 months from 1st March 2019 to 30th April 2019. We collected it from our project site at Harali, Osmanabad, Maharashtra using an IoT circuit system. The IoT circuit system was mainly divided into two parts:

- Master Device
- Slave Device

The workflow and basic working of these devices are described as follows:

1. Master Devices. The main objective of the Master device was to collect data of temperature, weight, and humidity from slave devices and then send that collected data to the smartphone using a Global network.[6] To collect the data from the master, the site person has to go to the main site and collect data in his smartphone from the master. The master sends/saves the records sent by the slave along with the received timestamp and its battery status on each record. Working of master devices is as follows:

Working. The main components of the master device are as follows:

- Special wakeup/sleep circuitry
- Microcontroller chip
- Transceiver circuit
- Microcontroller with inbuilt WIFI capability

Using the specialized circuit inside master devices, the circuit switches on at an interval of two hours from its sleep mode and gives power to the transceiver circuit inside master devices. The transceiver circuit then starts its receiver part for 10 minutes to check if any device is sending a battery low signal and if such a signal is found it stores the address of that device in its memory. After a 10 minutes interval, the microcontroller chip then takes one address from its memory, then the transceiver circuit using its transmitter starts transmitting the device address to the beehive boxes in an encoded format and waits for a positive acknowledgment from that specified beehive box. If a positive acknowledgment is not received within a few minutes of transmission, the master then treats it as a negative acknowledgment and stores it as an error along with an address that was being transmitted in its memory. If a positive acknowledgment is received, the transceiver circuit then starts receiving encoded data from that particular beehive box and stores its data into its memory.[8] This process is carried out to date from all the slave devices received. After receiving is done the special circuitry then turns off the transceiver part of the master device and turns on the microcontroller with WIFI capability. Microcontroller with WIFI capability then attempts to connect with a global network, in the meantime microcontroller chip then assembles the data and checks if there is any box with battery low signal and errors values in its memory and keeps it ready for transmission. After this process is done, the microcontroller then waits for an acknowledgment from the microcontroller with WIFI if it has connected to a global network or not. If a microcontroller is connected to a global network, then our microcontroller chip starts sending its assembled data to the microcontroller with WIFI, which in turn starts transmitting the data to mobile devices. After all, data is transmitted to the special circuitry then puts both the microcontrollers into their sleep mode and this goes on till the battery reaches a critical level. After the battery goes to a critical level, the wakeup circuitry then only wakes up the microcontroller with WIFI capability which in turn alerts the user with a mobile device with battery critical message in his smartphone.[9]

After the master transmits data to the smartphone, the smartphone then checks for any critical battery message from both master and slave devices and alerts the user accordingly. Block diagram of our Master device is shown in Figure 1 as follows:

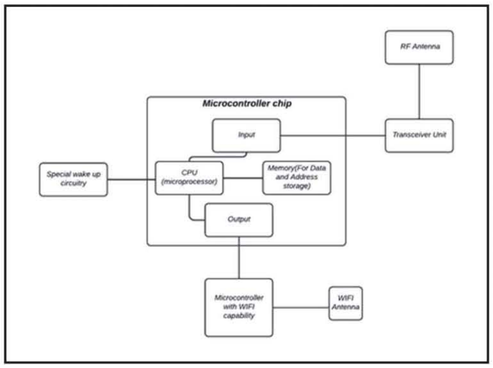

Fig. 1. Block Diagram of Master Device

2. Slave Devices. The main objective of slave devices was to collect data of temperature, weight, and humidity from beehive boxes and then send that collected data to the master device every two hours using high frequency(HF) and ultra-high frequency (UHF) radio waves transceiver circuits. This range was particularly chosen because these ranges possess no harm to the bees rather than the typical 2.4Ghz communication mediums.[7] The data was collected from 8:00 AM till 8:00 PM. The slaves send the device address, temperature, humidity, weight, battery percentage in an encoded form in an interval of two hours to its master. Working of individual slave devices is as follows:

Working. The main components of slave device consist of:

- Sensors (Weight, Humidity, Weight)
- Special wakeup/sleep circuitry
- Microcontroller Chip
- Transceiver Unit

Using the specialized circuit inside beehive boxes, the circuit switches on at an interval of two hours and using specialized sensors, collects data per minute continuously for 10 minutes. To get the most accurate value of temperature, humidity, and weight, the microcontroller chip inside the beehive box processes the data, encodes it, and keeps it ready to transmit it to our master device. [10] After collection and encoding of the data are done, our special circuitry then wakes up our transceiver circuitry. The receiving part of the transceiver circuitry then starts receiving encoded address from our master device and transfers that collected data from the master to our microcontroller chip, microcontroller then checks if the received encoded address is the same as that of its stored address and if the address matches then the transmitter part of the transceiver circuit first sends a positive acknowledgment to master and then starts transmitting data to the master system.[11] After all data is transmitted, the receiver part waits for a positive or negative acknowledgment from the master device. If it is a negative acknowledgment then the transmitter again sends the data to the transmitter and waits for an acknowledgment, above steps are followed till a positive acknowledgment is received from the master device. After positive acknowledgment is received, the special wakeup/sleep circuitry then turns off the transceiver part and then puts the microcontroller chip into its sleep mode for the next two hours and this goes on till the battery reaches a critical level. After the battery goes to a critical level, the wakeup circuitry then only wakes up the transceiver circuit which in turn alerts the master device with a battery critical message. Block diagram of our Slave device is shown in Figure 2 as follows:

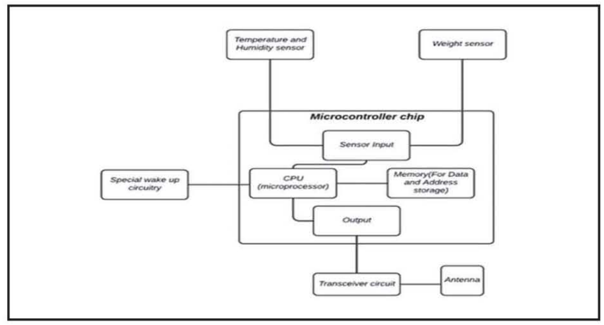

Fig. 2. Block Diagram of Slave Device

B. Attribute Information

TABLE I. TABLE OF ATTRIBUTES

S.no	Attribute	Description
1	Timestamp (dd/mm/yy/ hh:mm:ss)	Time at which data was recorded
2	Temperature (°C)	Temperature inside beehive
3	Humidity (%)	Humidity inside the beehive
4	Weight (kg)	Weight of the beehive

C. Data Analysis

1. Univariate Analysis:

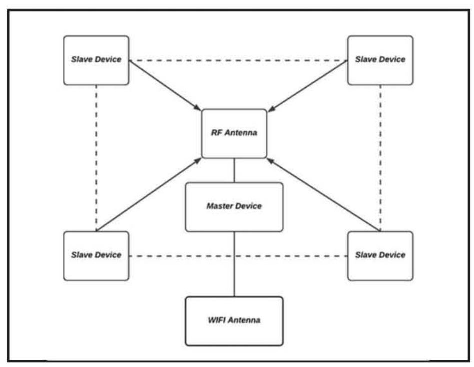

Fig. 3. Block Diagram of Overall Data collection system

Humidity. Upon visual inspection, humidity is high in the morning and evening, but it reduces significantly in the afternoon at around 2 pm.[12] The humidity of brood nests is extremely important since numerous studies have demonstrated that either high or low levels of humidity can affect the brood nests. Based on experimentations the relative humidity of beehives should not fall below 50% or else the eggs won't hatch. As shown in figure 4, data is collected in March and April, and hence it is less likely that the eggs were hatched during this period since the internal humidity did not cross 78.6% for a continuous period. It is likely to see such humidity fluctuations in the nest during brood-less periods.

The mean value of humidity for the observed beehive is approximately 29% since during daytime bees tend to store nectar which requires lower humidity levels. Another factor affecting humidity is the uncapped brood which is directly proportional to humidity. This is because the development of larvae and eggs is highly sensitive to desiccation.

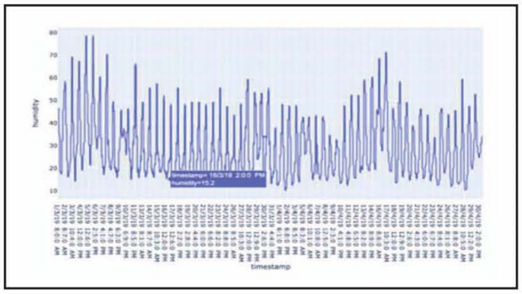

Fig. 4. Time Series plot of Humidity

Temperature: The average temperature recorded for April and May is approximately 33°C. Figure 5 displays the temperature distribution throughout March and April. It is observed that the temperature of the beehive rises to a great extent in the period of 12-4 pm. It goes above the threshold value of 35°C which is not optimal for brood nests.[13] Though they can resist up to 40°C, the internal structure of wax may start melting, affecting their formations and they may abscond if the temperature crosses the threshold value of beehive. After this threshold value is crossed, the bees beat their wings to renew the hive's atmosphere. The highest temperature recorded was 42.7° C. This temperature was recorded twice on 27th and 28th April consecutively at around 4 pm. This meant additional precautionary cooling measures by beekeepers and bees themselves.

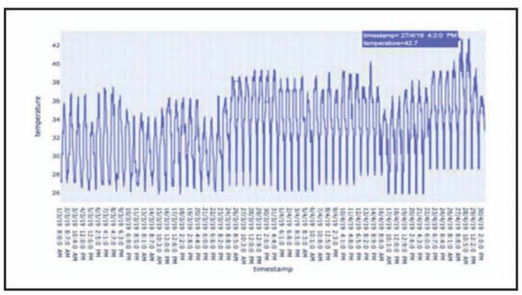

Fig. 5. Time Series plot of Temperature

Weight: We first started recording the weight on the first day when the honeybees first entered the beehive. The mean weight of beehives when they were emptied was 5.3 kg. However, it is important to note that the weight of the beehive includes the weight of the box which lies between 2 and 2.5 kg. During the period of observation, we noticed that the weight usually used to spike around the afternoon before decreasing a bit and then steadily climbing during the evening. As shown in figure 6, the maximum weight of the beehive was recorded as 6.64 kg on 31st March around 8 pm. We harvested the yield twice during March and only once during April. While it took only two weeks in March for the honeybees to produce the yield, in April it took the honey bees closer to three weeks to produce the same amount of yield.

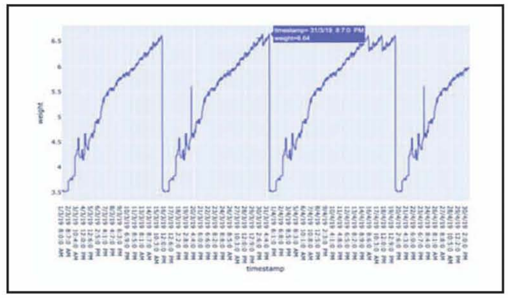

Fig. 6. Time Series plot of Weight

2. BIVARIATE ANALYSIS:

We calculated the correlation between temperature and humidity, between temperature and weight, and between humidity and weight using both Spearman's rank correlation and Pearson linear method. Using Spearman's rank correlation, we find that Rs(Temperature, Humidity) = -0.711, Rs(Temperature, Weight) = 0.083, Rs(Humidity, Weight) = 0.085. This indicates that there is a strong negative correlation between temperature and humidity, a weak positive correlation between temperature and weight, and a weak positive correlation between humidity and weight. Pearson's correlation on the same dataset is similar to spearman's rank correlation (Rp = -0.714, 0.087, 0.057 respectively).[14] Based on our previous inferences from the univariate analysis, we can infer that in the morning the temperature is low and hence humidity is high whereas the temperature increases as the sun reaches the peak and humidity decreases. [17] Univariate and bivariate analysis suggests that bees spend more time in the afternoon as the temperature is high. As the yield develops the space between the frame reduces and the bee colony becomes congested which raises the humidity in-side the beehive. This also reduces space for air circulation inside the hive and hence the ventilation is blocked. This can melt the yield. So, after performing an overall analysis we conclude that there is a necessity for controlled temperature and humidity inside the hive which can make the place suitable for bees to survive.

D. Logic Flow for the System

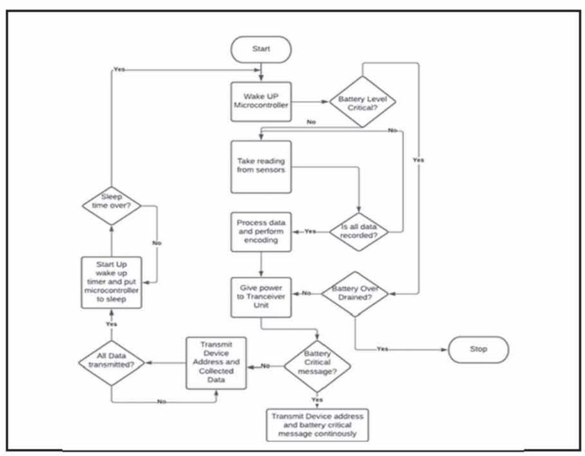

Fig. 7. Logic Flow of Slave device

At the start when the slave device wakes the Microcontroller, then it checks the status of the battery and if everything is fine, it proceeds to transmit the device address and all collected data. [16] It then activates a start-up timer and puts the microcontroller back to sleep. In cases where battery levels are low, it sends battery critical messages continuously to the master device.

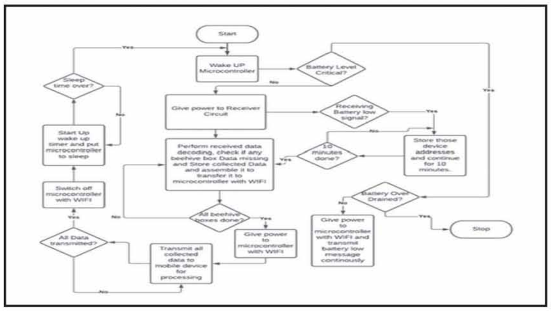

Fig. 8. Logic Flow of Master devices

At the start, the master device wakes up the microcontroller and checks the battery status, on presence of sufficient battery, it gives power to the receiver circuit which collects data from all beehives and decodes it before checking for missing values. It then assembles the data to transfer the data to the microcontroller with Wi-Fi which transmits all collected data to a mobile device for processing.[15] After the completion of the transmission, it activates the start-up timer, switches off the microcontroller with Wi-Fi capability and then puts the microcontroller to sleep. In case of low battery messages from any beehive, it stores the address of that beehive in the microcontroller itself and transmits it to the mobile device.

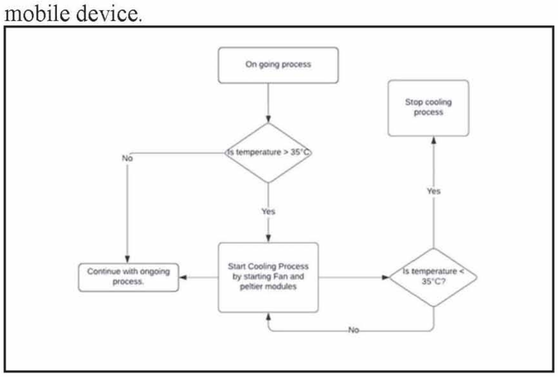

Fig. 9. Logic for automation

To automate the cooling process in the beehive, the sensors constantly check whether the temperatures are going above 35 C and start the fan and Peltier modules when the temperatures go above 35 C till the time that the temperature comes back down.

IV. Conclusion

The use of in-hive weather conditions and temperature and humidity sensors can be used for smart decision-making by farmers in India. IoT can further reduce the task of putting a wet gunny bag on each beehive by automating the cooling process. Peltier chips and fans can be used for cooling purposes at regular intervals. The overall system helps to protect the ecosystem and develop the yield quicker without com-promising the quality of the yield. Moreover, it helps bees to have the best conditions for reproduction. Temperature and humidity are the most important factors which are affecting the business of IoT. Due to the problem, bees migrate to another place and don't return to their hive. As a result, it leads to a big loss for the beekeeper. As this system helps beekeepers monitor the weight, it will not disturb the bees in the beehive.

We suggest that this whole research and analysis will help increase the productivity of beehives and motivate other farmers to practice beekeeping. This will also increase the number of beekeepers and beekeeping societies in India. This will not only benefit the beekeepers but also the farmers in a nearby location as bees play an important role in pollination. Further in the future predictive analysis can be done on the collect-ed data which will provide beekeepers an in-depth understanding of how much yield can be generated in a year. All this can be accomplished with the help of a beekeeper-centric mobile application.

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